## (a) One multi-angle QAOA cost layer, as a downstream compiler sees it

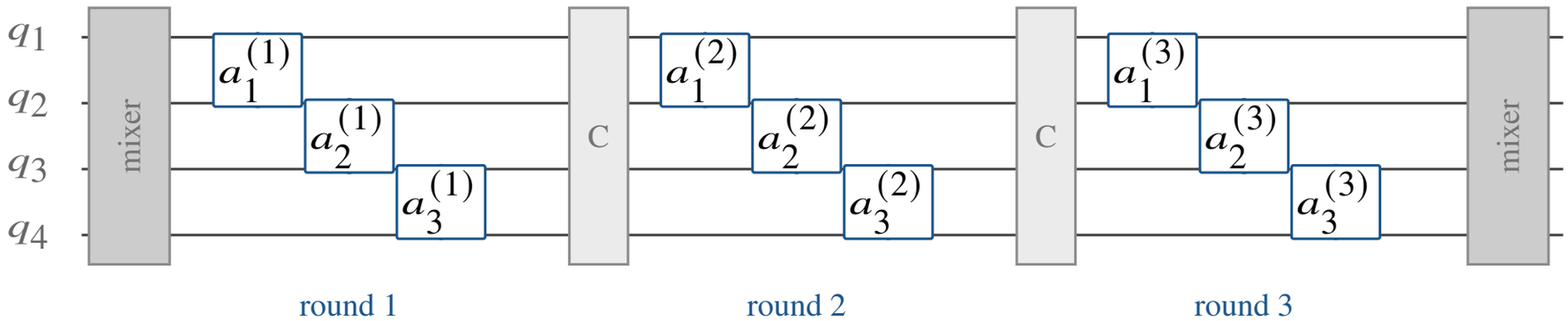
semantic $\exp\left(-i \sum_j \gamma_j Z_{u_j} Z_{v_j}\right)$

one record $(Z_{u_j} Z_{v_j},\ \gamma_j)$ per edge

one $R_{ZZ}$ per edge

schedule + twirl + lower

one contiguous commuting block, digitized into $r = 3$ rounds

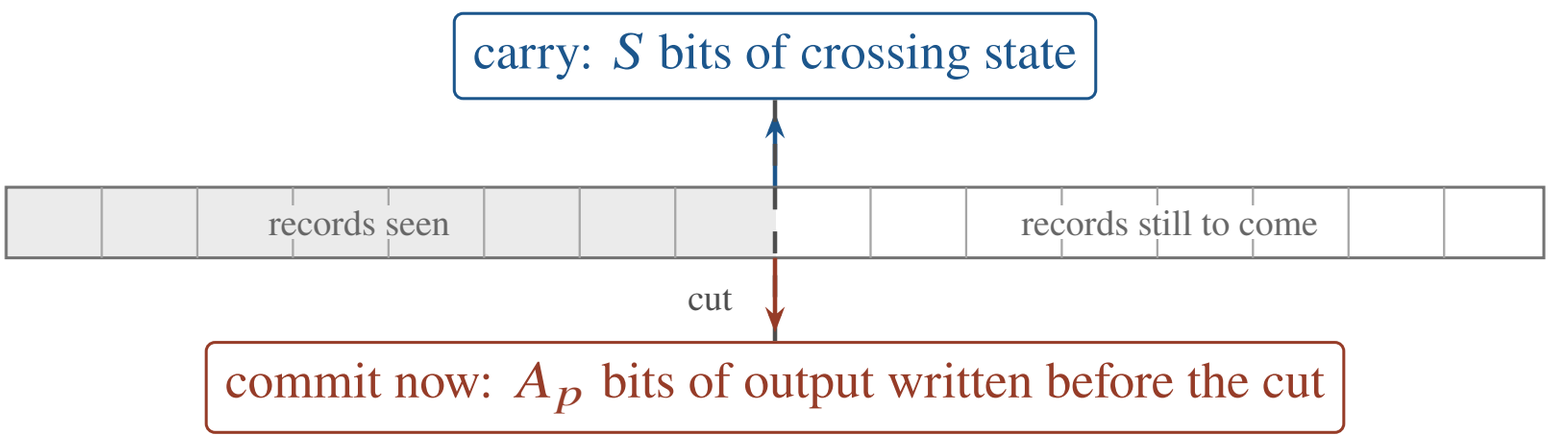



**dispersed:** $rm$ records reach the compiler. No record, and no prefix before the last round, determines the edge total $x_j = \sum_t s_j^{(t)} a_j^{(t)} \bmod Q$.

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contrast $\boxed{\text{c-}U^{2^0}}$ $\boxed{\text{c-}U^{2^1}}$ $\boxed{\text{c-}U^{2^2}}$ ✕ contiguous, but round 1 already fixes $\theta_j\,(2^r - 1)$

## (b) Two currencies, at one cut




$$\overline{A}_p + p\,S \;\geq\; (1-\delta)\,m \log_2 K_\epsilon - h_2(\delta) - \lceil \log_2(p+1) \rceil$$

plus a record-level $\log_2 \binom{m}{k}$ toll for executed (not buffered) commitment of a random mask